07 LECTURE

HEALTH CARE

MT 1, Second Semester LECTURE 07 - Ms. Megan Ruth C. Fernandez, RMT

VECTOR AND ZOONOSIS, CONTROL OF AIR, RADIATION, AND NOISE POLLUTION

TOPIC OVERVIEW

I. ZOONOSIS

- Any disease of infection that is **naturally transmissible from animals to humans**. (WHO, 2025)
- Can be caused by:
  - Bacteria
  - Viruses
  - Parasites
  - Fungi
- May involve unconventional agents
- Can spread to humans through direct contact or through food, water, or the environment
- It's important for us to study zoonosis because we have a close relationship with animals through agriculture and pets.
  - Because of this, the transmission of disease is also faster and we are more prone to disease carried by animals

Classification of Zoonosis

- Etiological Agents
  - Causative Agents
- Transmission Cycle
- Reservoir Hosts

A. CLASSIFICATION BASED ON ETIOLOGICAL AGENTS

BACTERIAL

Brucellosis

- ***Brucella***
- From **unpasteurized dairy products** or **infected animal tissues**
- Either through direct contact or accidental ingestion
- Could through transmitted through aerosols

Leptospirosis

- ***Leptospira interrogans***
- Could be transmitted through direct contact of **urine from rats or dogs.**
- Maintained in the **kidneys of wild and domestic animals**
- Just like rabies, not all dogs or rats have *leptospira interrogans*, but they may get it from the environment

Listeriosis

- ***Listeria monocytogenes***
- **Foodborne** - Cheeses, Raw or Unpasteurized Milk
- Animals could also get this disease and cause septicemia for them
- Mild manifestations for listeriosis include gastrointestinal issues

- Could cause spontaneous abortions in pregnant women
- Easily contracted by immunocompromised people

Tuberculosis

- ***Mycobacterium bovis* (cattles to humans)**
- *Mycobacterium Tuberculosis* (human to human)
- Transmitted through ingestion. Inhalation, or aerosols

Scrub Typhus | Rickettsia tsutsugamushi

- ***Oriental tsutsugamushi***
- Same family with *rickettsia rickettsii*
- Uses vectors
  - **Bites of larval trombiculid mites** (chiggers)

RMSF | Rocky Mountain Spotted Fever

- ***Rickettsia rickettsii***
- Bacterial infection, but uses vectors
  - **Spread by tick bites**

VIRAL

Rabies

- **Rabies virus (RABV)**
- Direct contact (usually saliva) through **bites or scratches** (depending on the viral load)
- Can manifest in a month to a year
- Not limited to cats and dogs, it also includes bats and raccoons.

Japanese Encephalitis

- **Mosquito borne flavivirus**
- **Culex spp**, endemic to Asia and Western Pacific
- Same family with the Dengue virus (Flavivirus)

Dengue

- **Flaviviridae family**
- **Aedes aegypti**
- Endemic to the Philippines

MYCOTIC OR FUNGAL

- Doesn't usually use fomites. Most of them are transmitted through aerosols

Dermatophytosis

- ***Microsporum and Trichophyton , Microsporum canis*** (cats and dogs)
- Superficial infection
  - Infection is only until the skin, hair, and nails
- Transmitted through beddings, floormates, grooming tools (especially cats and dogs)
  - Significant concern for veterinarians and animal handlers
- Only manifestation is a round rash redness

Cryptococcosis

- ***Cryptococcus neoformans* spores**
- Fungal Infection from **soil decaying wood and pigeon droppings**

Histoplasmosis

- ***Histoplasma capsulatum***
- Found in soil enriched with **bird or bat droppings**

PARASITIC

Toxoplasmosis

- ***Toxoplasma gondii***
- **Foodborne**
- Contact with **contaminated cat poop or eating contaminated food**.
- Can cause miscarriage in women
- More severe in immunocompromised people

Visceral larva migrans

- ***Toxocara spp.***
- **Dirt contaminated with pet poop** (cats and dogs)

Hydatid Disease

- larval stage of ***Echinococcus granulosus* tapeworms**
- Accidental ingestion of these tapeworms, eggs will appear in the liver, lungs, and brain.

B. CLASSIFICATION BASED ON TRANSMISSION CYCLE

ORTHOZOONOSES (DIRECT ZOONOSES)

- **Single vertebrate species**, either direct or indirect contact
- **Ex.** Anthrax, Rabies, Tuberculosis, Brucellosis

- **Ex. Anthrax** (*Bacillus anthracis*)

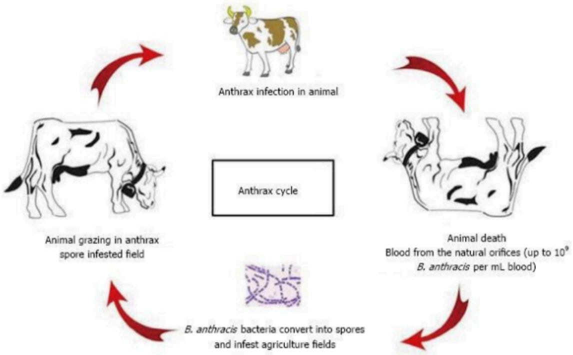

Figure 1. Anthrax Cycle

- In the whole lifespan of *Bacillus anthracis,* they may only infect 1 species
- Could also be transmitted through aerosols
- Also used as a bioterrorism agent
    - Has severe and fast-acting effects

CYCLOZOONOSES

- **Require 2 or more vertebrate hosts to complete transmission cycle**
- **Ex.** Taenia solium and saginata, Hydatid disease, Toxoplasmosis
- **Ex.** *Taenia solium*

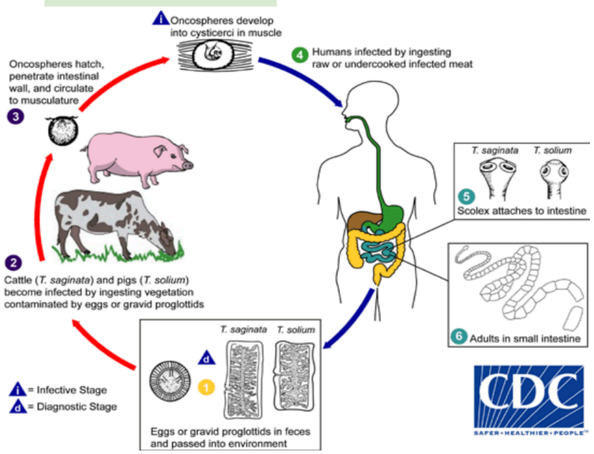

Figure 2. *Taenia solium* Cycle

- Contracted when consuming raw or uncooked meat (specifically pork)
    - Cysticerci goes to the small intestine and starts to grow as an adult
    - Needs to go to **another host** to grow

METAZOONOSES

- **Both Vertebrate and Invertebrate species** are involved in transmission.
- **Ex.** Yellow Fever, Paragonimiasis, Clonorchiasis
- **Ex.** Yellow Fever

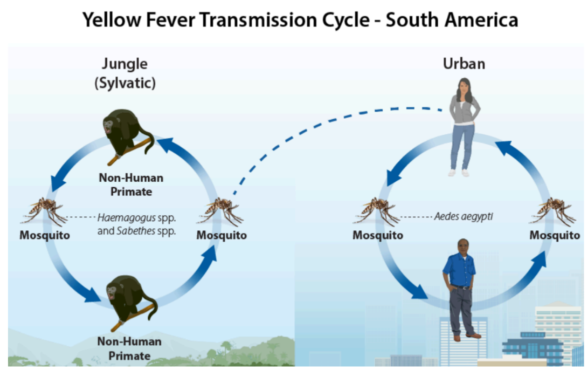

Figure 3. Yellow Fever Cycle

- A viral disease
- Even though the virus can live in a non-human primate, **it cannot infect humans unless they use vectors/invertebrate species** (mosquitoes)
- Other humans cannot infect other humans without an invertebrate vector

SAPROZOONOSES

- Require a **non-animate substance** in addition to vertebrate or invertebrate host.
- **Ex.** Cutaneous larva migrans, Histoplasmosis, Fascioliasis
- **Ex.** Cutaneous Larva Migrans

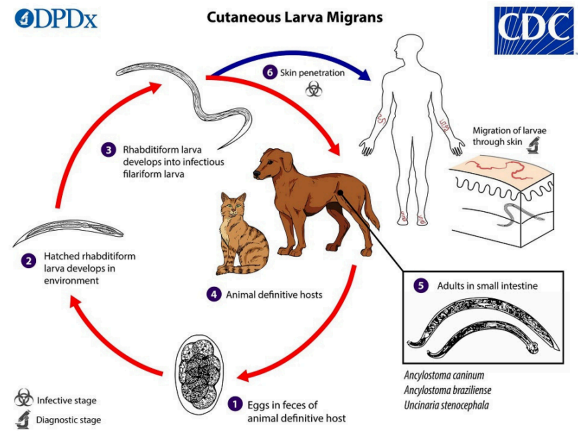

Figure 4. Cutaneous Larva Migrans Cycle

- Caused by hookworms (*Ancylostoma sp.)*
- One hookworm defecate in the feces of cats and dogs, it will stay in the environment
- In the soil of the environment (the non-animate substance), it will hatch and form into larva
- It will develop into an infected larva (filariform larva), which will then penetrate through the person's skin and infect
- **Uses feces and soil in their development**

C. CLASSIFICATION BASED ON RESERVOIR HOSTS

- **Reservoir Host**
    - Where the disease came from and where it is passed

Anthrapozoonoses

- Animals to Humans
- **Ex.** Rabies, Dengue

Zooanthroponoses (Reverse Zoonoses)

- Humans to Animals

Amphixenoses

- Man to Animal or Animal to Man
- **Ex.** *Staphylococcus aureus*, specially the MRSA (Methicillin-resistant Staphylococcus aureus) Type

D. CLASSES OF ZOONOSIS ACCORDING TO WHO

Endemic Zoonoses

- Present in many places and affect many people and animals.
- Expected to be many already
- **Ex.** Rabies, Brucellosis, Leptospirosis

Epidemic Zoonoses

- Sporadic in temporal and spatial distribution
- Not expected to increase
- **Ex.** COVID-19 (early months), Influenza, Plague

Emerging Zoonoses

- Newly appearing in a population or have existed previously but are rapidly increasing in incidence or geographical range.
- **Ex.** Rift Valley fever, SARS, pandemic influenza H1N1 2009, Yellow fever, Avian Influenza (H5N1) and (H7N9)

EMERGING ZOONOSIS

- Either **a really new disease** or **an old disease that reemerge** (due to change of habitat and population)
- Emerging zoonoses may be caused by **previously unknown agents** (i.e. 'true' emergence) or by **changes in the exposure or susceptibility of hosts** leading to an increased number of cases.
- Could be a global emergence or a local
- Some authors also distinguish between global emergence – outbreaks due to agents observed for the first time worldwide, or local emergence – occurrence of a known agent in geographical areas previously free of disease.

Factors Influencing Emerging Zoonoses

- Environmental Change
    - Climate Change & Deforestation
    - Fruit bats carrying Nipah Virus relocate near residential areas due to disturbance in habitat
- Human and Animal Demography (Statistics)
    - Overpopulation & Global Travel
- Pathogen Changes
    - Mutations made them stronger
        - Influenza Viruses (H1N1, H2N2, H2N3, etc.)
        - COVID-19 (Delta, Omicron)
- Changes in Farming Practice
    - Industrialized Farming to make up with the demands
        - Hygiene and sanitation is not taken into account anymore
- Social and Cultural factors
    - Food Preparation & Cultural Practices

E. TRANSMISSION ROUTES OF ZOONOTIC DISEASES

AEROSOLS

- Occurs when **droplets are passed through the air** from an infected animal and are breathed in by a person
- **Ex.** Droplets from tissues, soil contaminated with feces or urine and dust particles
- **Ex.** Tuberculosis

ORAL

- Occurs by **ingesting food or water contaminated with pathogen**
- **Ex.** Animal products such as milk and meat not pasteurized and cooked properly, eating or drinking after handling animals w/o washing hands
- **Ex.** Salmonella, Brucellosis

DIRECT CONTACT

- Coming into **contact with the saliva, blood, urine, mucus, feces, or other body fluids** of an infected animal
- **Ex.** Rabies through saliva, HIV through blood, Leptospirosis through urine

FOMITE

- An inanimate object that can carry a pathogen from an **animal to a person or from a person to another.**
- Usually a **fungal infection**
    - Because fungi sticks to surfaces, especially if they are moist and damp

- **Ex.** Door knobs, Contaminated brushes, Needles, Clothes

VECTOR-BORNE

- Being bitten by a tick, or an insect like a mosquito or a flea
- **Ex.** Dengue hemorrhagic fever transmitted by an infected mosquito

F. WORST PANDEMICS OF ZOONOTIC ORIGIN

The Black Death

- ***Yersinia pestis***
- Humans usually become infected through the **bite of an infected rodent flea** or by **handling an infected animal**
- Could also be human to human through **pustules, blood, or pus**
- Became a pandemic because trading ships carried this disease and lack of proper sanitation
- Occurs naturally in areas of western United States, **circulates among rodents and other animals**

1918 Flu Pandemic | *Spanish Flu*

- **H1N1 influenza virus**
- Lasting from **January 1918 to December 1920** (almost 3 years)
- **Infected 500 million people** (third of the population that time **estimated death toll of 17m to 50m**)
    - The reason why it was so deadly is because of the lack of vaccines, the ongoing war, and the population was not ready to handle a pandemic

NOTE

- Despite its name, the first case of the Spanish Flu was found in a military camp in the US during WWI, however, they did not report the outbreak out of fear of causing further panic. During the time, Spain was a neutral country and was the first to report the pandemic, which leads the rest of the world to believe that the pandemic started in Spain.

Asian Flu of 1957

- **Influenza A subtype H2N2**
- Death toll: 2M worldwide
    - The great decrease in death toll was due to vaccination for the virus
    - The subtype also did not target immunocompromised patients, rather, **the virus commonly targeted younger populations**, who have better immune systems.
        - 40% of deaths were immunocompromised or elderly patients
- First identified in East Asia

Hong Kong Flu | *1968 Flu Pandemic*

- **Influenza A (H3N2)**
- Death toll: 1M worldwide
- **Originated in China** in July 1968 - 1969-70
- **Still in circulation today** and is considered to be a **strain of seasonal influenza**

NOTE

- Avian flus are easily turned into pandemics because the migration of birds helps in spreading the disease.
- Other than that, viruses constantly mutate and gets stronger, despite the vaccines in place

HIV/AIDS Pandemic | *1920-Present*

- Still considered zoonotic because of its origin
    - Source of HIV infection in humans is believed to have come from a type of chimpanzee in Central Africa
- **75.7 million people have already been infected**
- Death toll: 32 M
- **Has no advertised definitive cure** but there are Antiretroviral drugs
    - Lowers the viral load to the point where it is undetected
    - So effective that the virus cannot be detected when taking the drugs

NOTE

- This disease blew up in the United States around the 1980s.
- Due to heavy misinformation, the original name for HIV in the United States was **Gay-related Immune Deficiency (GRID)** because they thought that only gay people could contract HIV.
- This information was quickly refuted because drug injection users and straight males contracted HIV years after.

COVID-19

- Caused by **Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-COV-2)**
- Possessions of anti-SARS-CoV antibodies in Rhinolophus affinis bats in China
    - **Bat was the most probable natural reservoir** or the primary host owing to the origin of the SARS-CoV-2.

Nipah Virus

- Cases of Nipah virus infection were **first reported in 1998**.

- The case fatality rate is estimated at 40% to 75%.
- **Fruit bats of the Pteropodidae family** are the natural host of Nipah virus.
- Nipah virus usually transmits from infected bats and other animals to humans and can also be transmitted directly between people.
  - To contract the virus, there needs to be **prolonged direct contact** or eating the fruits that an infected bat has bitten.

G. PRINCIPLES OF ZOONOSES PREVENTION, CONTROL, AND ERADICATION

- **Prevention**
  - **Inhibiting the introduction of a disease agent** into an area, a specific population group, or an individual
- **Control Efforts**
  - Steps taken **to reduce disease problem** to a tolerable level and maintain it at the level
- **Eradication**
  - **Final step** in disease control program
  - Consists of **complete elimination of disease** producing agent worldwide

- The basic principles of zoonoses prevention, control, and eradication programs are focused upon **breaking the chain of transmission** at its epidemiologically weakest link.
- Three factors involved:
  - **The reservoir**
  - **Transmission from the reservoir to the susceptible host**
  - **The susceptible hosts**

Reservoir Neutralization

- Ultimate source of zoonotic infection is the **infected reservoir host**
  - Through isolation or quarantine
- Manipulating the environment
  - Sanitation and Hygiene
- Three methods used to neutralize the reservoir are:
  - Removing infected individual
  - Rendering infected individuals or individuals as "non shedder" or not shedding the virus or organism to the environment
  - Manipulating the environment

Reducing Contact Potential

- Three methods are used…
  - Isolation and treatment of cases
  - Quarantine of possible infected individuals
  - Population control
    - Contact Tracing
- Remember!
  - **Isolation**: to keep the agent in
  - **Quarantine**: to keep the agent out

Increasing Host Resistance

- **Increase or boost immune system**
- Reduction of stress, by providing improved shelter and nutrition is not only an end but also a means to reduce the ravages of epidemics by increasing the survival ability of the affected population
  - Eating healthy, taking vitamins, exercise and getting enough sleep helps boost the immune system, thereby increasing the resistance of the host.

Chemoprophylaxis

- **Administration of a drug or medication** to prevent the development of a disease or infection or reduce severity
  - To decrease viral load
- Usually **post-exposure treatment**

Immunization

- Two purposes of vaccines:
  - To protect susceptible individuals from infection or disease
  - To prevent transmission of infectious agents by creating an immune population

H. ACTIONS IN ZOONOSIS CONTROL

- The following operational policies have been recognized as suitable and effective methods of zoonosis control:

Surveillance

- Testing of accessible animals and biological materials
  - Proper information of their manifestations
- Slaughterhouse surveys
- Isolation and typing of the zoonotic agent

Control in Animals

- Quarantine
- Test and Culling
  - As humanely as possible, if there are infected animals, they need to be slaughtered
- Test and Segregation
  - Separate infected animals for a period of time

- Immunization
    - For pets, poultry, and agriculture.
- Medical Treatment
- Movement Restrictions
    - Prevents animals from getting harmful diseases outside the environment
- Population Control

Prevention in Man

- Occupational Health Education
- Vaccination
- Post-exposure Treatment
- Proper food hygiene
- It's important for a certain area to not overcrowd because its makes the spread of disease faster and hygiene may not be maintained

II. VECTORS

Vector

- Vectors are **living organisms that can transmit infectious pathogens between humans, or from animals to humans**.
- Most of them are **insects or blood-sucking vectors**

How Do They Transmit Diseases?

- **Biological Transmission**
  Bites, Urine, Saliva, or Feces
- **Mechanical Transmission**
  Physical Contact
  In contact with their body, **especially feet**

A. VECTOR-BORNE DISEASES

- **Vector-borne diseases** are human illnesses transmitted by vectors.
- Account for more than 17% of all infectious diseases, causing more than 700,000 deaths annually.

B. MAIN VECTORS AND THE DISEASE THEY TRANSMIT

Vectors	Disease
Aedes sp.	Dengue Fever, Chikugunya, Rift Valley, Yellow Fever, Zika Virus
Anopheles sp.	Malaria (Parasitic Infection)
Culex sp.	Japanese Encephalitis, Lymphatic Filariasis, West Nile Fever
Sandflies	Leishmaniasis, Sandfly Fever
Ticks	Crimean-Congo Hemorrhagic Fever, Lyme Disease, Relapsing Fever, Rickettsial Disease, Tick-borne Encephalitis, Tularemia
Triatomine Bugs	Chagas Disease (American Trypanosomiasis)
Tsetse Flies	Sleeping Sickness (African Trypanosimiasis)
Fleas	Plague, Rickettsiosis
Black Flies	Onchocerciasis (River Blindness)
Aquatic Snails	Schistosomiasis (Bilharziasis)

Table 1. Main Vectors and the Disease they Transmit

NOTE

- Aedes sp., Anopheles sp., and Culex sp. are mosquito species
- Dengue is present in **132** countries and is very endemic in the Philippines
- When you are bitten by an Ixodes ticks (specific tick for Lyme disease), a common manifestation is a bulls-eye rash (Erythyma migrans).
- Triatomine (Kissing) Bugs usually bite when you are sleeping and bite in your lip area or eyes, and then they defecate or urinate and will cause the Chagas disease.
- Chagas disease can also be transmitted through blood transfusion, and vertical transmission
- Schistoma spp. lives in freshwater snails and when it is in its infective stage, it gets out of the snail.

C. CONTROL OF VECTORS

1. Hygiene and Control of Environment
2. Destruction of Pathogenic Material
    a. **Ex.** Incineration and Autoclaving
3. Disinfection
4. Food Hygiene

D. METHODS OF VECTOR CONTROL

Habitat Control

- **Removing or reducing the number of places where vectors can breed** helps to limit populations from growing excessively.
- **Ex.** Removing stagnant water to prevent Dengue and Malaria

Biological Control

- The **use of predators, bacterial toxins, or botanical compounds** can help control vector populations.
    - **Copepods** are released in bodies of water that have mosquito eggs
    - There are **larvivorous fishes** that eat mosquito eggs
        - Used to combat Malaria

Chemical Control

- Insecticides, larvicides (Baygon), and repellents are used to control pests and can be used to control vectors.
    - Effective is vectors are adults
- By doing chemical control we:
    - Avoid the risk of biting and contamination.
    - Limit the growth of populations at an early stage.
    - Dealing with adult populations.

E. VECTOR CONTROL MEASURES

1. Insecticide Treatment
2. House spraying
3. On-stream seeding
    a. Copepods and Larvivorous Fishes
4. On stream clearing
    a. Vegetations near streams are cleaned

F. ENHANCED 4S STRATEGY IN DENGUE PREVENTION

- Because of the Dengue outbreak in the Philippines, the DOH enhanced the 4S Strategy:

1. Search and Destroy
2. Seek Early Consultation
3. Self-Protection Measures
4. Say yes to fogging **only during outbreaks**
    a. Could be harmful to the environment and humans, and may cause pollution

III. AIR, NOISE, RADIATION POLLUTION

Pollution

- When substances or energy (e.g., light and heat) are introduced to the natural environment in **large amounts/concentrations** that can **cause harm** to humans, animals, and plants
- Most often composed of **synthetic, or human-made substances** (e.g., plastic)
    - Carbon dioxide (CO_2), sediment, and nutrients **can become pollutants if they exceed a particular level**
- Any substance could cause pollution **if they are in excess** and because of this excess, they can harm your humans, animals, or plants.

Pollutants

- A waste material that pollutes air, water, or soil
- Divides into 2 aspects: **natural and human-driven**
    - **Natural:** volcanic eruptions
    - **Human-driven:** Improper waste disposal

A. AIR POLLUTION

- "Contamination of the indoor or outdoor environment by any chemical, physical, or biological agent that modifies the natural characteristics of the atmosphere." (WHO)

PRIMARY SOURCES OF AIR POLLUTION

- Household air pollution
- Ambient (outdoor) air pollution
- Electricity in health-care facilities
    - Hospitals are required to have generators and they are fueled by diesel that has harmful damage to the air
- Sand and dust storms
- Automobiles
- Industries
- Tobacco smoke
- Pesticide spraying

AIR POLLUTANTS

- More than 100 air pollutants have been identified:
    - Carbon monoxide (CO)
    - Carbon dioxide (CO_2)
    - Hydrogen sulfide (H_2S)
    - Sulfur dioxide (SO_2)
    - Sulfur trioxide (SO_3)
    - Nitrogen oxides (NO_x)
        - shorthand for nitric oxide (NO) and nitrogen dioxide (NO_2)
    - Fluorine compounds
    - Organic compounds

EFFECTS OF AIR POLLUTION

Immediate Effects

- Damages the respiratory system and results in **acute bronchitis**

- May result in **immediate death by suffocation** if there is intense air pollution

Delayed Effects

- Smog can **damage the lungs and irritate the eyes and throat**
  - Smog is usually found in cities and happens when there is an accumulation of pollution.
  - When it is very thick, it can trap chemicals
- Benzene (in gasoline), a carcinogen (according to the EPA), can cause eye, skin, and lung irritation
- Chronic bronchitis, Lung cancer, Bronchial asthma, and Respiratory allergies

Socioeconomic Effects

- **Damages the buildings** (due to the corrosive nature)
- Kills plant and animal life (either by fucking up the soil or lessening the oxygen)
- **Reduces visibility** in urban towns compared to provinces

PREVENTION AND CONTROL OF AIR POLLUTION

- The WHO has recommended the following procedures for the prevention and control of air pollution

Containment

- Preventing the escape of toxic substances into the outdoor (ambient) air
- Usually engineering methods
  - Air ducts and vents

Replacement

- Replacing the process that causes air pollution with a process that doesn't cause air pollution
- Replacement of coal as a fuel source which reduces the smoke produces

Dilution

- Reduction of the air pollutant concentration by mixing it with sufficient air to render its presence unobjectionable
  - Ex: Placing greenery in under-construction buildings to lessen their effects on the atmosphere
- Only valid if it is within the self-cleaning capacity of the environment

Legislation

- Should be hand in hand with implementation
- **RA 8749 - The Philippine Clean Air Act**
- **Comprehensive Air Quality Management Policy and Program**

International Action

- **WHO** established an **international network** of laboratories that **monitor and study** air pollution

DISINFECTION OF AIR

Mechanical Ventilation

- To **control indoor air quality, excess humidity, odors, and contaminants**

Ultraviolet Radiation

- Germicidal UV light can destroy or **deactivate the genetic material of the microorganisms when exposed to air surfaces**

Chemical Mists

- Air disinfectants (usually chemical substances) **disinfect microorganisms suspended in the air**

Dust Control

- To **reduce surface and air movement of dust**

B. NOISE POLLUTION

- High sound levels may lead to adverse effects in humans or other living organisms
- Sound levels **less than 70dB** are not **damaging** to living organisms, **no matter how long or consistent** the exposure is (WHO)
- More than 8 hours of constant noise that is **beyond 85 dB** may be **hazardous**

NOISE LEVELS

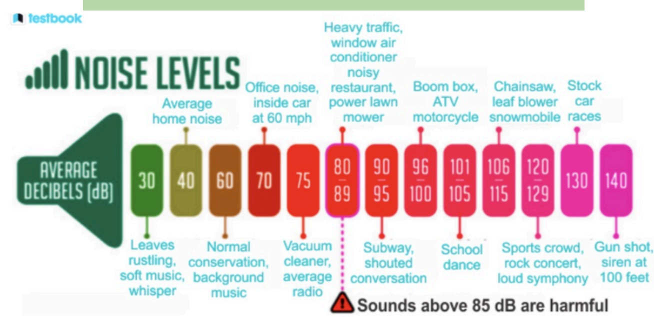

Figure 5. Noise Levels

- Isn't only from one source; **noise accumulates**

SOURCES OF NOISE POLLUTION

- Domestic noises from radios, TV sets, and transistors
- Acute noise levels near railway junctions, airports, traffic roundabouts, and bus terminals
- **Festivities**, particularly at night, with loudspeakers in full volume and pressure horns (i.e., recreational use)

HEALTH EFFECTS OF NOISE POLLUTION

- Hypertension
    - Increased stress could lead to constricted vessels
- Hearing Loss
- Sleep Disturbances
- Child Development
- Psychological Dysfunctions and Noise Annoyance

- It has been proven that we are irritable and anxious when it's noisy and it's not a suitable environment for a child to grow up in

CONTROL OF NOISE

- Careful planning of cities
- Control of vehicles
- Industry and railways outside the city
- Protection of the exposed person
- Legislation
- Education

C. RADIATION POLLUTION

Radiation

- Energy that comes from a source & travels through some material or through space
- Part of a man's environment

Radioactive Pollution

- **Increase** in natural radiation levels due to human activities
- About **20%** of the radiation we are exposed to is because of human activities

SOURCES OF RADIATION

Natural

- Cosmic Rays
- Environmental
    - Terrestrial, Atmosphere

Man-Made

- Medical and Dental
- Occupational exposure
- Nuclear
- Miscellaneous
    - TV sets. Radioactive dial watches. Isotope-tagged products, and Luminous markers

TYPES OF RADIATION

- Ionizing radiation is the type of radiation that **penetrates the skin** could stay there
- Ionizing Radiation can be divided into 2 types:
    - Electromagnetic Radiations
        - X-rays and gamma rays
    - Corpuscular Radiations
        - Alpha and beta particles

DETERMINANTS OF BIOLOGICAL EFFECTS

- Rate of Absorption
    - Amount of radiation entering the body
- Area exposed
- Variations in species and individual sensitivity
    - Cockroaches are 10x more resistant to radiation
- Variations in cell sensitivity

APPEARANCE OF BIOLOGICAL EFFECTS

Acute Effects

- Effects **seen immediately** after **large doses** of radiation are delivered over a **short time**
    - Ex: Radiation sicknesses and burns

Delayed Effects

- Appears **months or years** after radiation exposure
    - Ex: Cataract formation and cancer induction

BIOLOGICAL EFFECTS

Acute Localized Effects

- Most common effects are **skin-related**
    - **200 rad:** mild erythema
    - **300 rad:** temporary hair loss
    - **600 rad:** more severe erythema
    - **700 rad:** permanent hair loss
    - **1000 rad:** desquamation in tissue and necrosis

Long-Term Effects

- Experience delayed effects due to **acute high-dose exposures** or from **chronic long-term exposures** over many years
    - Carcinogenic
    - Embryological
    - Cataractogenic
    - Life span shortening

Genetic Effects

- Inheritable mutations in DNA
- Seen in mammals (*animals), yet no convincing evidence in humans
- Difficulty in measuring because of the ff.:
    - Subtle effects
    - Long life spans
    - Uncertainties in the background rate
    - Confounding factors
- **77,000 births** from **Japanese bomb survivors** show **no substantial evidence** of genetic effects

HOW CAN RADIATION EXPOSURE BE REDUCED?

- **Less time** spent near the source
- **Greater distance** from the source
- Behind **shielding** from the source
    - Workers must wear a **film badge or dosimeter**
        - **Film badge:** a device that **measures and records** exposure to ionizing radiation

Figure 6. Film Badge

IV. REFERENCES

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